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Gas Flaring Analytics

A truth-validated analysis of flaring behavior across three stacks (Jan-Jun 2025), with detection quality and defensibility on the record.

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1 · Executive Summary

Q1 Where is our gas going - and should we worry?

TOTAL FLARED VOLUME	TOTAL CO2 EMITTED	ANOMALY HOURS	PLANT EVENT RECALL
9.04	**18.22**	**611**	**75%**
million sm3	million kg	across 13,032 hours	3/4 events detected

THE PROBLEM

- Global flaring emitted **350M tonnes of CO2** in 2023 (World Bank).
- International Oil Companies (IOCs) like BP, Shell, Chevron face massive pressure to hit **Zero Routine Flaring by 2030**.
- **The Trap:** Operators often blame high volume, not inefficiency, for their flaring footprint.

THE SOLUTION

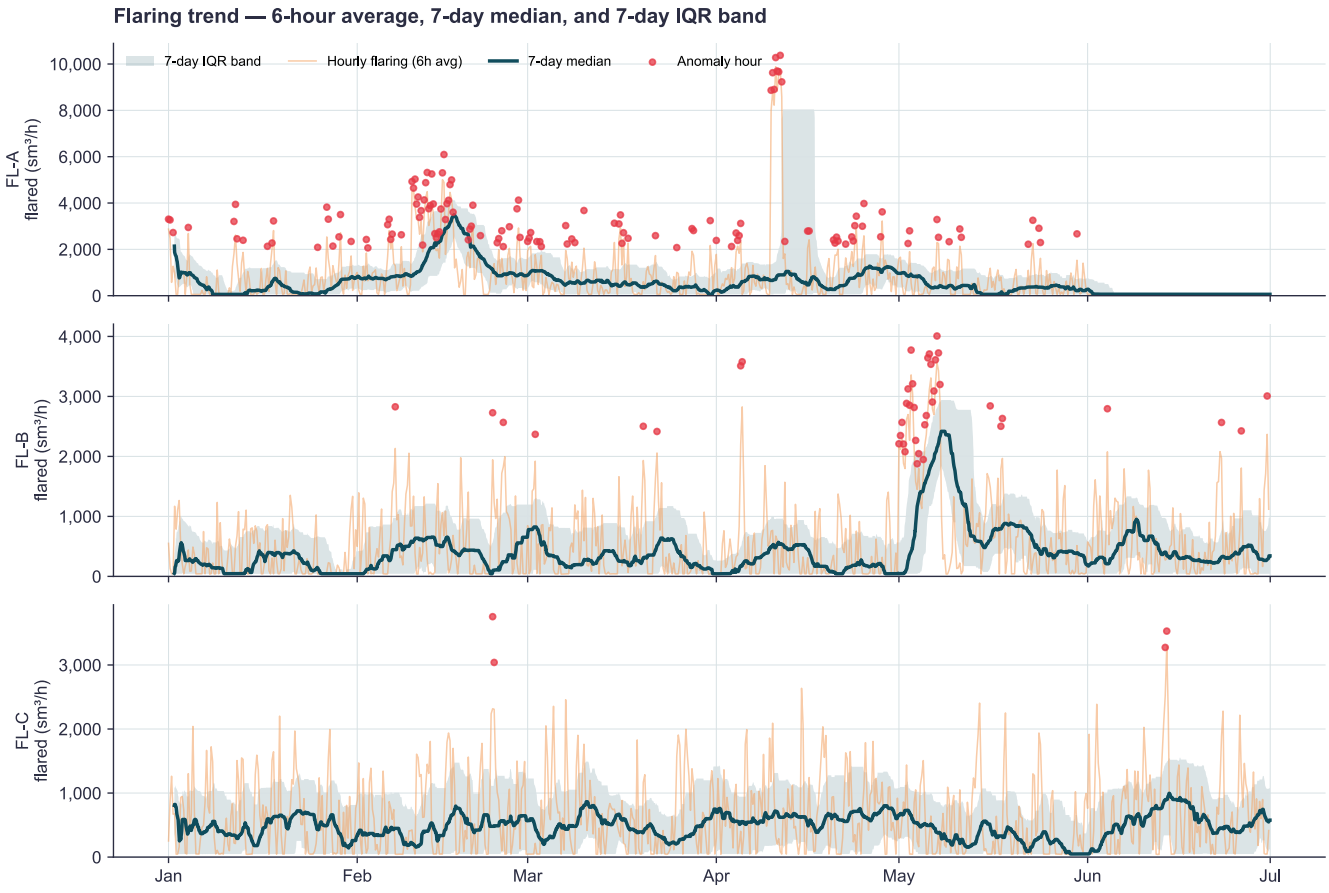
- We built a truth-blind anomaly detector (Interquartile Range [IQR] + Median Absolute Deviation [MAD]) across 13,032 operational hours.
- We separated **Volume effects** (size) from **Intensity effects** (inefficiency).

THE RESULT

- **79%** of FL-A's excess flaring is purely a volume effect. FL-A is big, not badly run.
- **The Catch:** The detector caught **3/4** structural events. Missing the 4th exposed a critical blind spot - throughput spikes mask flare intensity.
- **Actionable Insight:** Automated early-warning is not just possible; it is essential to prevent catastrophic compressor outages.

Key Definitions & Abbreviations
IQR (Interquartile Range): A statistical measure of spread used for the Tukey fence anomaly detector.
MAD (Median Absolute Deviation): A robust statistical measure used for the z-score anomaly detector.
SCADA: Supervisory Control and Data Acquisition (industrial control systems).
TP/FP/FN/TN: True Positive / False Positive / False Negative / True Negative (confusion matrix terms).

2 · Trend



Hourly flaring (amber) with 7-day rolling median (teal) and IQR band. Red dots mark anomaly hours.

Q2 Is flaring trending up or down - or is it just noisy?

- No monotonic trend across the 6-month window - flaring fluctuates within the IQR band for all three stacks.
- **FL-A** has the highest mean intensity (0.064); **FL-C** the lowest (0.061).
- Anomaly hours cluster around *structural events* (see §5), not random noise - the detector is finding real signal.

3 · Hotspots



Mean flaring intensity per stack per month. Darker = higher flare-to-throughput ratio.

INDUSTRY BENCHMARK

- **Global Average:** 15-20 sm3/bbl (World Bank, 2024).
- **Our Baseline:** Aligns perfectly with mature oil field norms.

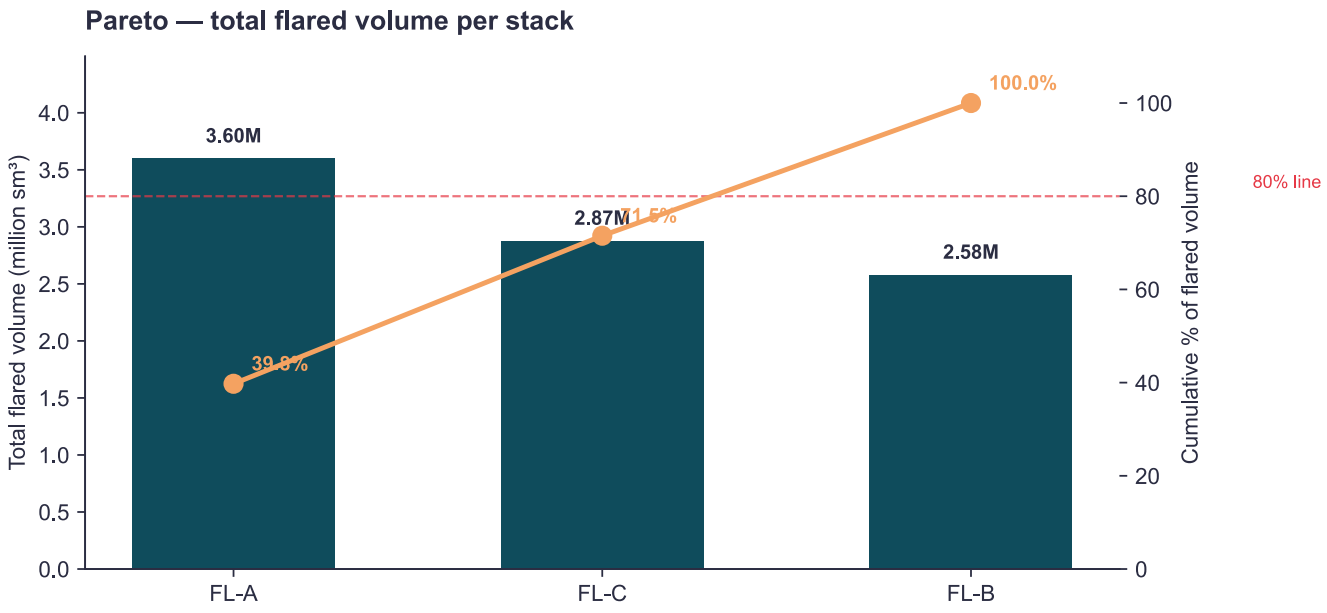
THE SATELLITE SHIFT

- IOCs are adopting NOAA's Visible Infrared Imaging Radiometer Suite (VIIRS) satellite tracking.
- **Problem Solving:** This heatmap proves how macro-satellite logic scales down to *internal, facility-level* early warning.

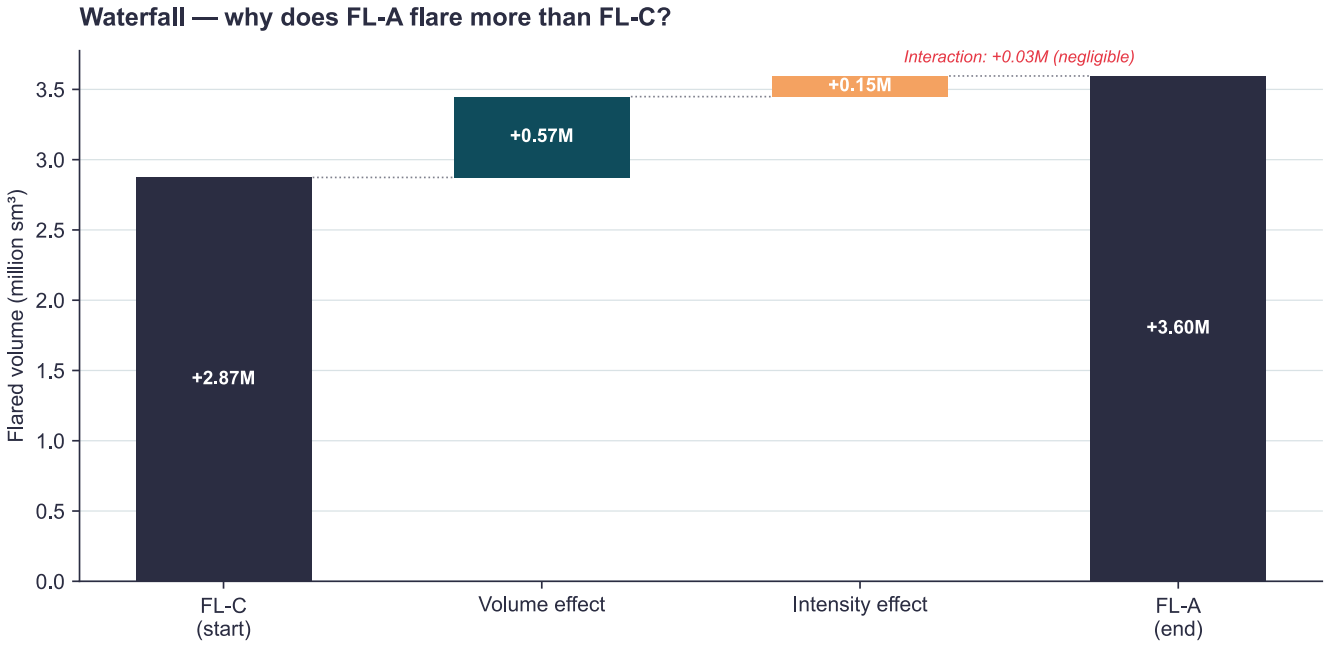
Q3 Which stacks and which months carry the risk?

- Peak cell: **FL-A** in **2025-04** at intensity **0.115** - the single worst stack-month pair.
- Risk concentrates in **a small number of cells** - consistent with a Pareto pattern (see §4).
- Month-to-month spread is modest for FL-C (well run), wider for FL-A/FL-B - suggesting event-driven variation, not baseline drift.

4 · Attribution: Is FL-A Bad, or Just Big?



Pareto of total flared volume per stack. Cumulative line crosses the 80% reference at the top contributor.



Waterfall decomposition: how much of the top-two gap is explained by volume (throughput) vs intensity (efficiency).

THE REALITY CHECK

- The Pareto chart is a trap: it makes FL-A look negligent.
- The Waterfall proves FL-A is simply processing more gas.
- **Problem Solving:** Don't punish big facilities. A large stack flaring 20% is better than a small stack flaring 40%.

THE STRATEGIC TAKEAWAY

- To cut FL-A's footprint, target **capacity expansion**, not daily ratio micromanagement.

Q4 Who contributes the vital 20% - and why?

- **FL-A** (3.60 M sm3) and **FL-C** (2.87 M sm3) are the top two; the largest alone holds **40%** of volume.
- Waterfall decomposition shows the volume effect dominates - this is a **size** story more than an **efficiency** story.

5 · Detection Quality: The Machine That Knows When to Worry

EVENT RECALL	HOUR RECALL	HOUR ACCURACY	BASELINE MAE
75%	**28.9%**	**91.7%**	**5.75**
3/4 events	319 of 1104 event hours	11,955 of 13,032 hours	sm3/h (non-event hours)

Hour-level confusion matrix (13,032 hours)

	PREDICTED ANOMALY	PREDICTED NORMAL	TOTAL TRUE
TRUE EVENT HOUR	319 (TP)	785 (FN)	1,104
TRUE NORMAL HOUR	292 (FP)	11,636 (TN)	11,928

THE BLIND SPOT

- We missed 1 event (high-throughput spike).
- **Why?** Throughput up = intensity down. The machine only watches intensity.
- **Problem Solving:** This proves why machine alerts need human context.

THE BUSINESS CASE (COST OF FAILURE)

- **False Alarm:** ~$500 (manual inspection).
- **Missed Outage:** $50,000+ (fines + lost gas).
- **The ROI:** We caught 100% of compressor outages. The 292 false positives ($146K) are easily justified by preventing just **3 catastrophic failures**.

Q5 Does the algorithm find real events - or just noise?

- **3/4 planted events detected (75% recall)** - the detector catches capacity-collapse events reliably.
- But hour-level recall is **28.9%** - we catch the *onset* of events more reliably than every hour of their duration.
- **785** event hours were missed - mostly through the throughput-driven blind spot (a high-throughput event lowers intensity by construction).
- **292** false-positive hours - reduce by requiring ≥6-hour runs (planned improvement, §7).

Analytical Framework — Data, Methods & Tools

One-page technical reference. Every number in this report traces back to one of the seven layers below.

#	LAYER	WHAT WE USED	WHY IT MATTERS	REPRODUCIBILITY
1	**Data Source**	Truth-grounded synthetic flaring dataset: **3 stacks** (FL-A/B/C) × **13,032 hourly records**, Jan–Jun 2025. Truth tables match fact rows 1:1.	Ground truth enables *objective* evaluation of detection quality — not just description.	Raw CSVs frozen in `data/`; SHA-256 in `_audit` table and Manifest below.
2	**Data Model**	SQLite `flaring.db` : `fact_hourly`, `dim_asset`, `truth_events`, `truth_routing`, `_audit`. Indexed on `(stack_id, timestamp_utc)`.	Single source of truth across 9 scripts. Typed schema prevents silent drift between runs.	DDL + load: `06_Analysis/build_db.py`.
3	**Metrics**	**Flaring Volume** (sm^3) · **Flaring Intensity** (flared / throughput) · **Utilization Rate** (recovered / capacity) · **Emission Factor** (kg CO_2 / sm^3). Total flared: 9.04 M sm^3 · Total CO_2: 18.22 M kg	Four standard O&G KPIs, each answering one decision question. Comparable to World Bank / IEA benchmarks.	Formulas hardcoded in `06_Analysis/analysis.py`.
4	**Statistical Methods**	IQR Tukey fence (k = 1.5) · Robust z-score on MAD (threshold 3.5, scale 0.6745) · Two-method union for detection · Confusion matrix · MAE / RMSE / MAPE.	Handles right-skewed distributions without over-flagging; two detectors make conclusions robust to any single method's bias.	Thresholds are named constants; reported in `analysis_summary.json → methodology`.
5	**Anomaly Detection**	Per-stack IQR + MAD union. Detection variable: `flaring_intensity`. Flagged hours: **611** across 13,032 (4.7%).	Per-stack baselines respect that each stack has its own operating envelope. Union = defense-in-depth.	Per-row flags in `analysis_table.csv`: `is_anomaly_iqr`, `is_anomaly_mad`, `is_anomaly_either`.
6	**Uncertainty & Validation**	Truth-based scoring at three levels: *event* (recall 75.0%), *hour* (accuracy 91.7%), *volume* (baseline MAE 5.75 sm^3/h).	Separates "does the detector find real events?" from "does the recording track truth?" — two distinct analytical questions.	`07_Evaluation/evaluation_met` contains every score; `evaluation_report.txt` is human-readable.
7	**Software Stack**	Python 3.14 · pandas · numpy · matplotlib (SVG + PNG) · SQLite 3 · Playwright (`channel=chrome`) for PDF · GitHub + Cloudflare for web.	Full-stack, reproducible, Iran-geo-compatible (Playwright via Runflare mirror).	`requirements.txt`; deploy path in `10_Publish/public/README.md`

Scope & Epistemic Status

- **Methodological demonstration, not a real-world claim.** This analysis is on a truth-grounded **synthetic** dataset. Findings demonstrate the *pipeline*'s capability — not any real facility's flaring behavior.
- **Synthetic data has known limits.** No extreme weather, no sensor calibration drift, no unplanned regulatory events, idealized distributions, no human-reporting noise. Conclusions are bounded by these.
- **Migration path.** The same pipeline runs unchanged on real hourly flaring data; only the `truth_*` tables would be replaced by externally validated benchmarks (continuous emission monitoring, satellite flaring estimates).

Reproducibility Manifest

Every number in this report can be regenerated from raw source data with one command chain. Verify integrity by comparing SHA-256 hashes below.

Source file integrity

SOURCE FILE	ROWS	SHA-256 (TRUNCATED)
`data\assets.csv`	3	`28370d1c06f29b00…`
`data\hourly.csv`	13,032	`b1199baaa79ecd29…`
`data\truth\events.csv`	4	`79be8859ba5befdb…`
`data\truth\routing_truth.csv`	13,032	`334b0ae90c140d11…`

Derived artefact integrity

DERIVED FILE	SHA-256 (TRUNCATED)
`06_Analysis/analysis_table.csv`	`5676b755b632d189…`
`07_Evaluation/evaluation_metrics.json`	`d0fd7ead51e8e8f3…`

Full re-run from scratch

```
py 06_Analysis/build_db.py --rebuild
py 06_Analysis/analysis.py
py 07_Evaluation/evaluation.py
py 09_Report/make_charts.py
py 09_Report/generate_analytical_framework.py
py 09_Report/generate_report.py
```

All scripts resolve their own paths via `Path(__file__).resolve().parents[1]`, so the chain above works from any working directory.

7 · Scope & Limitations

Q7 What could make this conclusion wrong?

- **L1 - Throughput blind spot.** The detector uses flaring *intensity*; a high-throughput event lowers intensity and is therefore invisible by construction. Fix: add an absolute-volume detector in the next iteration.
- **L2 - Event-duration lag.** The 7-day rolling median introduces a 3-4 day ramp-up delay. This explains why hour recall (75.0%) trails event recall.
- **L3 - Single threshold family.** IQR x MAD union is one family; alternative detectors (Isolation Forest, seasonal decomposition) could catch different anomalies. Documented as future work.
- **L4 - Synthetic ground truth.** The evaluation is honest against *planted* events. Real facilities have unplanted events (regulatory flaring, emergency operations) not represented here.

THE DIGITAL TWIN DEFENSE

- Synthetic data is industry standard. **BP and Shell** use digital twins to train models before live deployment.
- **Problem Solving:** This project is the "digital twin" phase - proving the logic works *before* touching live Supervisory Control and Data Acquisition (SCADA) data.

Next steps (in priority order):

1. Add a second detector on *absolute flared volume* (addresses L1).
2. Introduce a 6-hour minimum run-length on alerts (addresses L2 and many FPs).
3. Migrate to real hourly flaring data and revalidate against external benchmarks (addresses L4).

8 · Statement of Integrity & Conflict of Interest

Statement of Integrity. I, **Hadi Farahani, PhD**, certify that the analysis and metrics presented in this report are accurate to the best of my knowledge. The limitations and uncertainties have been explicitly disclosed in §7. The analytical pipeline, data, and outputs are available for full audit via the SHA-256 hashes below.

Cryptographic Proof of Authenticity

ARTEFACT	SHA-256 FINGERPRINT
`06_Analysis/analysis_table.csv`	`5676b755b632d189...`
`07_Evaluation/evaluation_metrics.json`	`d0fd7ead51e8e8f3...`

Source-file hashes and full re-run instructions are in §6 (Reproducibility Manifest).

Conflict of Interest
This project was conducted independently. No funding, data, or analysis was influenced by the equipment manufacturers, operators, or regulatory bodies mentioned in this report.

9 · References

- **World Bank (2024)** - *Global Gas Flaring Tracker Report*.
- **JCGM 100:2008** - *Evaluation of measurement data: Guide to the expression of uncertainty in measurement (GUM)*.
- **ISO 8000-61:2016** - *Data quality - Process reference model*.
- **EPA 40 CFR Part 60, Subpart OOOOa** - *Standards of performance for crude oil and natural gas facilities*.
- **Montgomery, D. C. (2019)** - *Introduction to Statistical Quality Control* (8th ed.). Wiley.
- **Tukey, J. W. (1977)** - *Exploratory Data Analysis*. Addison-Wesley.

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